- Set operators combine the results of two or more SELECT queries.
- Oracle provides four main set operators to work with result sets.
  - UNION
  - UNION ALL
  - INTERSECT
  - MINUS

syntax

- SELECT column1, column2, ...
- FROM table1
- SET_OPERATOR
- SELECT column1, column2, ...
- FROM table2;

- SELECT emp_id, emp_name, department, salary
- FROM hr_employees
- UNION
- SELECT emp_id, emp_name, department, salary
- FROM it_employees
- UNION
- SELECT emp_id, emp_name, department, salary
- FROM sales_employees
- ORDER BY emp_id;

- SELECT emp_id, emp_name, department, salary, 'HR' as source
- FROM hr_employees
- UNION ALL
- SELECT emp_id, emp_name, department, salary, 'IT' as source
- FROM it_employees
- UNION ALL
- SELECT emp_id, emp_name, department, salary, 'Sales' as source
- FROM sales_employees
- ORDER BY emp_id;

INTERSECT - Employees who exist in multiple departments

- -- Employees in both HR and IT
- SELECT emp_id, emp_name
- FROM hr_employees
- INTERSECT
- SELECT emp_id, emp_name
- FROM it_employees;

- -- HR employees not in all_employees
- SELECT emp_id, emp_name, department
- FROM hr_employees
- MINUS
- SELECT emp_id, emp_name, department
- FROM all_employees;

What is an Index ?

- An **index** is a database object that improves the **speed of data retrieval** on a table.
- It works like the index of a book — instead of scanning every row (full table scan), Oracle uses the index to locate rows faster.

- select emp_id,emp_name,department from employees; --cjnvs28u7sj9v
- select * from v$sql;
- select * from v$sql_plan;
- select * from v$sql where sql_text like('select emp_id,emp_name,department from employees%');
- select * from v$sql_plan where sql_id='cjnvs28u7sj9v';

Types of Index

- B-Tree Index
- Bitmap Index
- Composite Index
- Unique Index
- Function Based Index
- Index Organized table

B-Tree Index

- Balanced Tree Index
- Its an ordered list of values that are divided in to range associated with rowid.
- It is recommended to create the index on high cardinality columns.
- **High cardinality** → many unique values

Bitmap Index

- It stores the data in the form of bitmap array, stores the index key column information along with your rowid.
- Its is recommended to create on low cardinality columns.

Composite Index

- Created on multiple columns in a table.
- It is also called as concatenated index.

Unique index

- Doesn’t allow duplicate values and allow null values

Function Based Index

- If an index created using function or expression the it is said to be function based index.
- It can be either B-Tree or Bitmap.

Index organized Table

- Organizing the data at the time of inserting.
- You must have primary key on the table.

View

- View isa logical/virtual table that doesn’t physically exist in the database and stored in oracle data dictionary.
- View stores SQL Statements/Queries in the database.
- It will not hold any data of its own instead it retrieves/fetches the data from the base table/view whenever its required.
- A view will looks and acts like a table.
- No data's is stored in view.

- Business Intelligence
- Need of Business Intelligence
- Terms used in BI
- Components of BI

- Business Intelligence is the process of taking raw data from your business and turning it into meaningful and useful information that enables better, data-driven decision-making.

- BI is a technology-driven process for analyzing data and presenting actionable information to help corporate executives, business managers, and other end users make informed business decisions.
- It encompasses a wide range of tools, applications, and methodologies that allow organizations to collect data from internal systems and external sources, prepare it for analysis, develop and run queries against that data, and create reports, dashboards, and data visualizations.

- Think of BI as the dashboard of a car. A car's dashboard takes raw data from the engine (RPM, speed, fuel level, engine temperature) and presents it to the driver in an easy-to-understand format.
- The driver doesn't need to know the intricate mechanics; they just need to see the speedometer to know if they're speeding or the fuel gauge to know when to refill.
- Similarly, BI takes complex business data and presents it in a way that managers can quickly understand what's happening and what to do next.

The Need for Business Intelligence

- **Informed Decision-Making:** This is the core need. BI replaces intuition with facts. Instead of guessing which marketing campaign worked best, BI can show you precisely which one generated the highest ROI.
- **Improved Operational Efficiency:** BI systems can identify bottlenecks and inefficiencies in processes. For example, a BI report might show that a particular stage in the supply chain consistently causes delays, allowing managers to address it directly.
- **Gain Competitive Advantage:** By understanding market trends, customer behavior, and their own performance better than their competitors, companies can spot opportunities and threats early and act swiftly.

The Need for Business Intelligence

- **Identify New Revenue Opportunities:** Analyzing sales and customer data can reveal untapped markets, potential for upselling/cross-selling, or new product ideas that management hadn't considered.
- **Understand Customer Behavior:** BI helps in segmenting customers, analyzing purchasing patterns, and measuring customer satisfaction. This leads to more effective marketing, improved customer service, and higher retention rates.
- **Track Key Performance Indicators (KPIs):** BI allows organizations to monitor their health in real-time by tracking KPIs like sales growth, customer acquisition cost, employee productivity, and profit margins on centralized dashboards.
- **Regulatory Compliance and Risk Management:** For many industries, reporting to regulatory bodies is mandatory. BI systems can streamline this process. They also help in identifying potential risks, such as a sudden drop in sales in a key region or a rising debtors' list.

Key Terms Used in BI

- Database
  - Relational
  - NOSQL
- **Data Warehouse:** A central repository where data from various operational systems (like sales, marketing, HR) is stored, cleaned, and integrated. It is designed specifically for query and analysis, not for transaction processing.
- **Data Mart:** A subset of a data warehouse, designed for a specific line of business (e.g., a data mart for the sales department or for finance). It's smaller and more focused.

Key Terms Used in BI

- **OLAP (Online Analytical Processing):** A technology that allows users to analyze multidimensional data from multiple perspectives. For example, a user can view sales data by *region*, by *time*, by *product*, and by *salesperson* all at once.
- **Dashboard:** A graphical user interface that provides an at-a-glance view of key performance indicators (KPIs) and metrics, often using charts, graphs, and gauges.
- **KPI (Key Performance Indicator):** A measurable value that demonstrates how effectively a company is achieving key business objectives. Examples: Monthly Recurring Revenue, Customer Churn Rate, Net Promoter Score (NPS).
- **Data Mining:** The process of discovering patterns, correlations, and anomalies in large datasets using machine learning, statistics, and database systems.

Key Terms Used in BI

- **Ad-hoc Reporting:** The process of creating custom, on-the-fly reports to answer a specific business question, without having to use a pre-built, standard report.
- **Big Data:** Extremely large and complex datasets that are difficult to process using traditional data processing applications. BI tools are increasingly incorporating big data analytics.
- **ELT(Extract Load and Transform)**
- **ETL (Extract, Transform, Load):** The three-step process that is fundamental to data warehousing.
  - **Extract:** Data is pulled from different source systems (e.g., CRM, ERP, Excel sheets).
  - **Transform:** Data is cleaned, standardized, deduplicated, and reformatted into a consistent structure.
  - **Load:** The transformed data is loaded into the data warehouse.

General concept of Data Warehouse

- Data Warehouse
- History of Data Warehousing
- Need for Data Warehouse
- Data Warehouse Architecture
- Data Mining Works with DWH
- Features of Data warehouse
- Data Mart
- Application Areas

Online Analytical Processing (OLAP)

- Online Analytical Processing (OLAP)
- Nature of OLAP analysis
- Types of OLAP
- OLAP Tools
- OLTP and OLAP Capgemini Public
- OLAP Functional requirements
- OLAP Fast and Selective
- Operational versus Informational System

Online Analytical Processing (OLAP)

- OLAP is a category of software technology that enables analysts, managers, and executives to gain insight into data through fast, consistent, interactive access to a wide variety of possible views of information.
- This information has been transformed from raw data to reflect the real-dimensional understanding of the enterprise.

Nature of OLAP Analysis

- **Slice:** Selecting one specific dimension from the cube, creating a sub-cube.
  - *Example:* "Show me all data for Year = 2023." This is like taking a single, flat slice through the time dimension.
- **Dice:** Selecting a specific sub-set of data across multiple dimensions.
  - *Example:* "Show me data for Product Category = 'Electronics' and Location = 'Europe' and Time = Q1 & Q2." You are carving out a smaller cube from the larger one.

Nature of OLAP Analysis

- **Drill Down/Up:** Navigating through levels of a dimension hierarchy.
  - **Drill Down:** Going from a summary to more detailed data.
    - *Example:* Year Total -> Quarterly Sales -> Monthly Sales.
  - **Drill Up (or Roll-Up):** The opposite. Going from detailed data to a summarized level.
    - *Example:* Sales by City -> Sales by State -> Sales by Country.
- **Pivot (or Rotate):** Changing the dimensional orientation of a report or view.
  - *Example:* Swapping the rows and columns in a report. What was on the rows (Products) moves to the columns, and what was on the columns (Time) moves to the rows. This provides a different perspective on the same data.

Types of OLAP

- Multidimensional OLAP (MOLAP)
  - The classic approach. Data is stored in a proprietary, pre-calculated, multidimensional format (the "cube").
- Relational OLAP(ROLAP)
  - Works directly with the relational database (the data warehouse). It uses a star or snowflake schema and dynamically generates SQL queries to perform calculations.
- Hybrid OLAP(HOLAP):
  - A combination of MOLAP and ROLAP. It stores aggregate data in a MOLAP cube for speed and keeps detailed data in the relational database.

OLAP Tools

- **Microsoft Power BI**
- **Tableau**
- **Qlik Sense**
- **Microsoft Analysis Services (SSAS)**
- **Oracle Essbase**
- **SAP BW/4HANA**

OLTP vs. OLAP

15 mins

- **OLAP Functional Requirements (The "CORE" Requirements)**
- **OLAP: Fast and Selective**
- **Operational vs. Informational System**

Data Mining

- Data Mining is the process of discovering hidden, previously unknown, and potentially useful patterns, correlations, and knowledge from large volumes of data.
- It goes beyond mere data observation or querying; it involves using sophisticated algorithms and statistical techniques to **predict trends and behaviors** and to **identify relationships** that are not immediately obvious.

The Knowledge Discovery Process (KDD)

- Data Mining is not a single step; it is a crucial part of the larger Knowledge Discovery in Databases (KDD) process. KDD is the overall process of converting raw data into useful knowledge.
- **Data Cleaning**
- **Data Integration**
- **Data Selection**
- **Data Transformation**
- **Data Mining**
- **Pattern Evaluation**
- **Knowledge Presentation**

The Need for Data Mining

- **Information Overload:** The volume of data is too massive for manual analysis.
- **Automated Pattern Discovery:** Humans are good at spotting patterns in 2 or 3 dimensions, but data exists in dozens of dimensions. Algorithms can find complex, multi-dimensional patterns we would never see.
- **Competitive Advantage:** The ability to predict customer behavior, optimize operations, and mitigate risks before competitors do is a massive business advantage.
- **Decision Support:** Moving from intuition-based to data-driven decision-making.

- Use of Data mining
- Data mining and Business Intelligence
- Types of data used in Data mining

Types of Data Used in Data Mining

- **Relational Databases:** The most common source, with tables, rows, and columns.
- **Data Warehouses:** The ideal source, as data is already cleaned and integrated.
- **Transactional Data:** Records of transactions (e.g., sales, logs).
- **Flat Files:** CSV, Excel spreadsheets.
- **Multimedia Data:** Images, audio, video.
- **Text Data (Text Mining):** Documents, emails, web pages, social media posts.
- **Web Data (Web Mining):** Clickstream data from web servers.
- **Time-Series Data:** Data points indexed in time order (e.g., stock prices, sensor data).

Data Mining Applications (By Industry)

- **Retail:**
  - Market Basket Analysis for store layout and promotions.
  - Customer segmentation for loyalty programs.
  - Demand forecasting for inventory management.
- **Banking & Finance:**
  - Credit scoring and risk analysis.
  - Fraud detection in credit card and insurance claims.
  - Algorithmic trading.
- **Telecommunications:**
  - Churn prediction to identify customers likely to switch providers.
  - Campaign optimization.

Data Mining Applications (By Industry)

- **Insurance:**
  - Claims analysis for fraud detection.
  - Customer profitability analysis.
- **Healthcare:**
  - Predicting disease outbreaks.
  - Identifying effective treatments for specific patient groups.
- **Manufacturing:**
  - Predictive maintenance for machinery.
  - Optimizing supply chains.

Data Mining Products

- **Standalone Open-Source Tools / Libraries:**
  - **R & R Libraries:** A powerful statistical programming language with packages like dplyr, caret, and randomForest for data mining.
  - **Python & Scikit-Learn:** The dominant language for data science, with Scikit-Learn being a premier library for machine learning. Also includes libraries like Pandas, NumPy, and TensorFlow.
  - **Weka:** A popular, free, GUI-based Java software suite for machine learning.

Data Mining Products

- **Integrated within Major BI Platforms:**
  - **Microsoft SQL Server Analysis Services (SSAS):** Includes powerful data mining algorithms.
  - **SAP BW/BI:** Integrated analytics capabilities.
  - **Oracle Data Mining:** Algorithms built into the Oracle Database.
- **Cloud-Based AI/ML Platforms:**
  - **Microsoft Azure Machine Learning**
  - **Google Cloud AI Platform**
  - **Amazon SageMaker**
    These are the modern trend, offering scalable, managed services for building, training, and deploying data mining models.

ETL: The Engine of Data Integration

- ETL stands for **Extract, Transform, Load**. It is a fundamental process in data warehousing and BI that involves:
- **Extracting** data from heterogeneous source systems.
- **Transforming** it into a clean, consistent, and useful format.
- **Loading** it into a target data warehouse or data mart.
- The primary goal of ETL is to consolidate disparate data into a single, trusted source of truth that is optimized for analysis and reporting.

The ETL Process in Detail

- **Stage 1: Extract**
- This is the phase of data acquisition from various source systems.
- **Sources:** Operational databases (e.g., Oracle, SQL Server, MySQL), CRM systems (e.g., Salesforce), ERP systems (e.g., SAP), flat files (CSV, Excel), web logs, and APIs.
- **Methods:**
  - **Full Extract:** The entire dataset is copied from the source each time. Simple but can be resource-intensive for large datasets.
  - **Incremental/Delta Extract:** Only the data that has changed since the last extraction is captured. This is more efficient and is common in production systems. It often uses timestamps, logs, or incrementing keys to identify changes.

Stage 2: Transform

- This is the most complex and critical stage where data is cleansed, standardized, and prepared for analysis. Common transformation tasks include:
- **Data Cleansing:** Correcting inconsistencies and errors (e.g., standardizing "NY," "New York," "N.Y." to a single value).
- **Data Validation:** Enforcing data integrity rules and rejecting/filtering invalid records.
- **Deduplication:** Identifying and removing duplicate records.
- **Format Standardization:** Converting data types and formats (e.g., converting MM/DD/YYYY to YYYY-MM-DD).
- **Calculations and Derivation:** Creating new calculated fields (e.g., Profit = Revenue - Cost).
- **Aggregation:** Summarizing data (e.g., daily sales totals to monthly totals).
- **Joining and Merging:** Combining data from multiple sources or tables.
- **Splitting Columns:** Breaking a single column into multiple ones (e.g., splitting "Full Name" into "First Name" and "Last Name").
- **Pivoting/Unpivoting:** Changing the structure of the data for better storage in the data warehouse.

Stage 3: Load

- This is the process of writing the transformed data into the target data warehouse.
- **Methods:**
  - **Full Load:** The entire target table is erased and reloaded with the new data. Used for small dimensions or during initial loads.
  - **Incremental Load:** Only new, changed, or updated records are inserted into the target table. This is the standard for large fact tables to maintain performance.
- **Frequency:** The load can be scheduled (e.g., nightly, weekly) or happen in near real-time (a process often called ELT or Change Data Capture).

Metadata Used in ETL

- **Source Metadata:**
  - Source system names, connection strings, and types (e.g., SQL Server, Oracle).
  - Table and column names from the source.
  - Data types and formats in the source.
- **Transformation Metadata (Business Rules):**
  - The logic for each transformation (e.g., "Trim whitespace from Customer_Name").
  - Data validation rules (e.g., "Email must contain an '@' symbol").
  - Mapping documents that define how source fields relate to target fields.
- **Process Execution Metadata:**
  - **Job Logs:** Start time, end time, and status (Success/Failure) of each ETL job.
  - **Performance Statistics:** Number of rows extracted/transformed/loaded, processing time for each step.
  - **Error Logs:** Details of any records that failed validation or processing, including the error reason. This is vital for data quality management.

Metadata in Data Warehousing

- **Technical Metadata:**
  - **Purpose:** Used by IT staff and database designers to manage and maintain the warehouse.
  - **Includes:**
    - Table structures, data types, and primary/foreign key relationships in the warehouse.
    - ETL process details (as described above).
    - Partitioning strategies, indexing schemes, and storage information.

Metadata in Data Warehousing

- **Business Metadata:**
  - **Purpose:** Used by business users to understand and trust the data they are analyzing.
  - **Includes:**
    - Business definitions of terms (e.g., "What is the exact business definition of an 'Active Customer'?").
    - Data owner information (who is responsible for the data).
    - Calculated field logic (e.g., "Profit is defined as Revenue minus Cost of Goods Sold and Marketing Expenses").
    - Report and dashboard descriptions.

Metadata in Data Warehousing

- **Operational Metadata:**
  - **Purpose:** Provides information about the operation and usage of the data warehouse.
  - **Includes:**
    - Data lineage: Shows the full flow of data from its source to its current form in a report. (e.g., "This KPI on the CEO's dashboard comes from this column in the fact table, which was loaded by this ETL job from this source system").
    - Data currency: When was the data last updated? Is this a live connection or a nightly snapshot?
    - Access patterns and performance statistics: Who is using what data and how often?